

Written exam 18 December, 2019

Course title: Network Optimization

Course no: 42115

Aids allowed: All

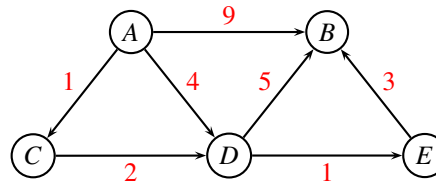
Exam duration: 4 hours

Weighting: Points are indicated on the assignments. A maximum of 44 points can be obtained.

Language: You can write in Danish or English.

Shortest path problem

Consider the following directed graph where distances w_{ij} are marked with red.



Assume that you should find the shortest path from A to all other nodes. If two solutions have the same length you would prefer the one visiting fewest edges.

Problem 1: (2 point) Describe how you would transform the distances, to make it possible to solve the problem using an ordinary shortest-path algorithm. Your transformation should work for *any* graph, not only the above. ■

Problem 2: (3 point) Solve the problem. State which algorithm you choose, and give sufficient details to make it possible to follow the algorithm. Report the shortest-path tree and optimal distances. ■

Critical path with time windows

Consider a critical path problem, in which each task i has an earliest start time e_i .

Problem 3: (2 point) Show how the critical path problem with start times can be transformed to an ordinary critical path problem. ■

Problem 4: (4 point) Solve the below problem:

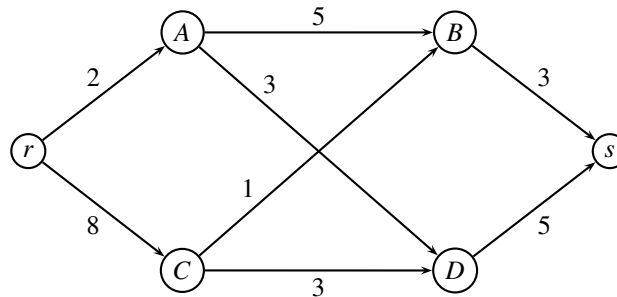
activity i	processing time p_i	predecessors	earliest start e_i
A	2	C	1
B	6		0
C	4		3
D	3	B, C	5

■

Problem 5: (2 point) Indicate the critical path, and argue how you found it. Will the critical path only be on tasks A, B, C, D or can it be on non-tasks? ■

Maximum flow problem

Consider the following maximum flow problem where the capacity u_{ij} is indicated on edge $(i, j) \in E$.



Problem 6: (5 point) Solve the problem using the *preflow-push* algorithm. (You can use the figures on the last page, and enclose them to the solution. Remember to write what you are doing.) ■

Problem 7: (2 point) Find a minimum cut and the corresponding capacity of the cut. (Explain how you find the cut). ■

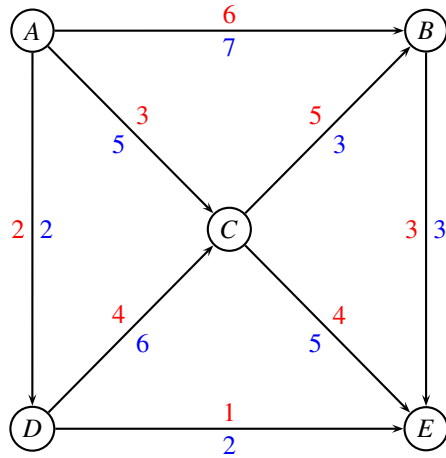
Workforce plan

A building company has to handle 3 projects, P1, P2, P3, over the next 4 months. Project P1 can start in the beginning of month 2, and must be completed at the end of month 3. Project P2 and P3 can start in the beginning of month 1, and must be completed, respectively, at the end of month 4 and 2. The projects require, respectively, 9, 10, and 11 man-months. Each month, 8 engineers are available. Due to the internal structure of the company, no more than 6 engineers can work, at the same time, on the same project. Determine whether it is possible to complete the three projects within the time constraints and, if it is possible, find a feasible workforce plan.

Problem 8: (5 point) Describe how this problem can be reduced to the problem of finding a maximum flow in an appropriate graph. (Hint: Look for a feasible flow of value 30.) ■

Repositioning containers

Consider the following liner shipping network where costs w_{ij} are indicated with red, and capacities u_{ij} (measured in TEU) are indicated with blue.



Due to imbalance in trade, some ports have a surplus of containers, while other ports are in demand of containers. The following table states the surplus for each port (negative values indicate a demand):

Port i	A	B	C	D	E
Surplus 20" containers σ_{1i}		-1		2	-1
Surplus 40" containers σ_{2i}	1	-1		1	-1
Surplus high-cube containers σ_{3i}	1				-1

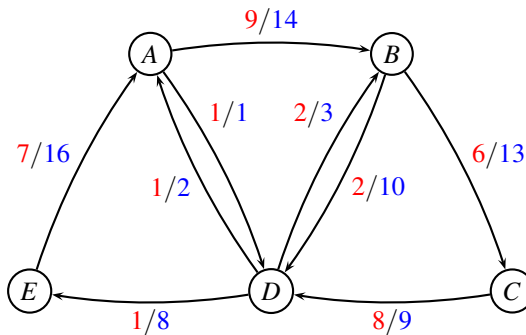
A 20" container takes up $w_1 = 1$ TEU space, a 40" container takes up $w_2 = 2$ TEU space, while a high-cube container takes up $w_3 = 4$ TEU space.

Problem 9: (2 point) We would like to re-balance the containers. Calculate the demand d_i in each port i measured in TEU. ■

Problem 10: (5 point) Solve the resulting minimum cost flow problem using demands d_i , and report the cost. ■

Multicommodity-flow problem

MEDITLINE (Route net of a Mediterranean liner shipping company)



For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij} (in TEU).

Note: The route net is the same as in Project Assignment 2019, except that edge (A, B) has capacity 14, edge (A, D) has capacity 1, edge (B, D) has capacity 10.

The capacity is measured in TEU (*Twenty-foot Equivalent Units*). A 20" container takes up $w_1 = 1$ TEU space, a 40" container takes up $w_2 = 2$ TEU space, while a high-cube container takes up $w_3 = 4$ TEU space. Containers may not be split. We have the demands

k	s_k	t_k	d_k	w_k
1	A	D	3	1
2	E	D	4	2
3	B	A	2	4

Commodity $k = 1$ is shipping three 20" containers ($w_k = 1$) from A to D, commodity $k = 2$ is shipping four 40" containers ($w_k = 2$) from E to D, and commodity $k = 3$ is shipping two 40" high-cube containers ($w_k = 4$) from B to A.

Consider the LP-relaxed path formulation of MEDITLINE, in which we have added the three dummy routes: AD , ED , BA of cost 100. To ensure a feasible solution, the dummy routes do not use any capacity in the existing network.

After a few iterations of column generation we have added routes ABD and AD for commodity 1, and $EABD$ for commodity 2. This leads to the following *restricted master problem*:

$$\begin{aligned}
\min \quad & 100x_0^1 + 100x_0^2 + 100x_0^3 + 11x_1 + 1x_2 + 36x_3 \\
\text{s.t.} \quad & \begin{aligned} & + 1x_1 + 2x_3 \leq 14 \quad (y_{AB}) \\ & + 1x_2 \leq 1 \quad (y_{AD}) \\ & + 1x_1 + 2x_3 \leq 13 \quad (y_{BC}) \\ & + 1x_1 + 2x_3 \leq 10 \quad (y_{BD}) \\ & \leq 9 \quad (y_{CD}) \\ & \leq 2 \quad (y_{DA}) \\ & \leq 3 \quad (y_{DB}) \\ & \leq 8 \quad (y_{DE}) \\ & + 2x_3 \leq 16 \quad (y_{EA}) \end{aligned} \\
& x_0^1 + x_1 + x_2 \geq 3 \quad (\bar{y}_1) \\
& x_0^2 + x_3 \geq 4 \quad (\bar{y}_2) \\
& x_0^3 \geq 2 \quad (\bar{y}_3) \\
& x_0^1, x_0^2, x_0^3, x_1, x_2, x_3 \geq 0
\end{aligned} \tag{1}$$

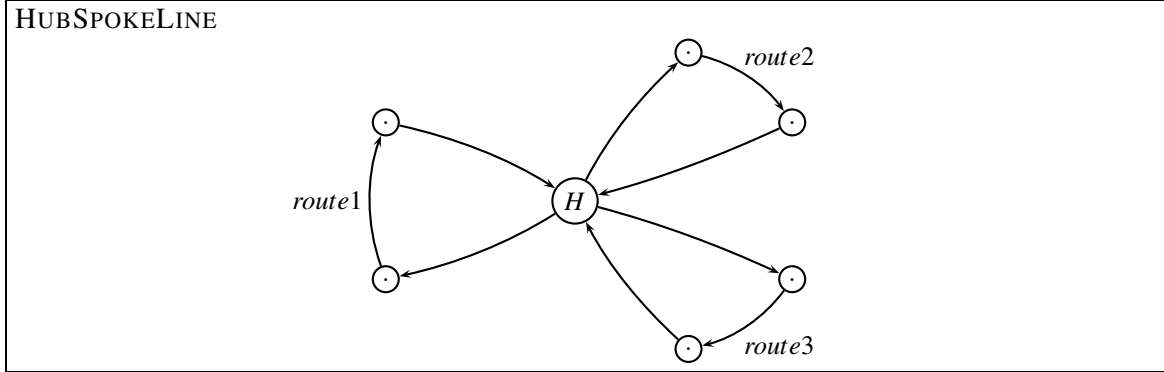
Let y_{ij} denote the dual variable corresponding to the capacity constraint for edge (i, j) , and let $\bar{y}_k$ denote the dual variable corresponding to the demand constraint for commodity k .

The dual variables for the above model (1) are $y_{AD} = -10$ while $y_{ij} = 0$ for all other edges (i, j) . Moreover, $\bar{y}_1 = 11$, $\bar{y}_2 = 36$, $\bar{y}_3 = 100$.

Problem 11: (5 point) Solve the pricing problem for each of the three commodities as a shortest-path problem. Show the graph, and give sufficient details to follow the calculations. ■

Problem 12: (2 point) Which column should be added to the restricted master problem, and why? ■

Modeling a hub port



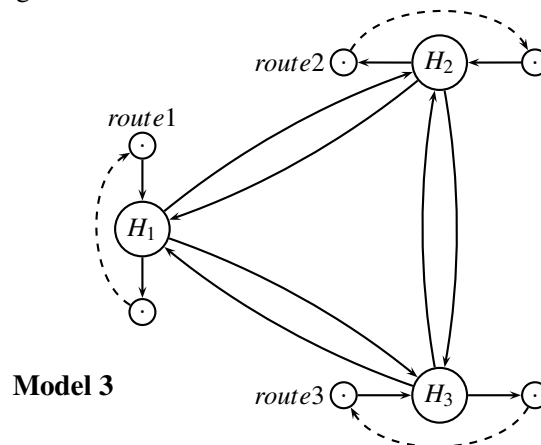
We will now study how the hub port H in a liner shipping network, HUBSPOKELINE, can be formulated in graph terms as part of a multicommodity flow network. We have three routes arriving to the hub as follows:

route	weekday
1	Wednesday
2	Wednesday
3	Friday

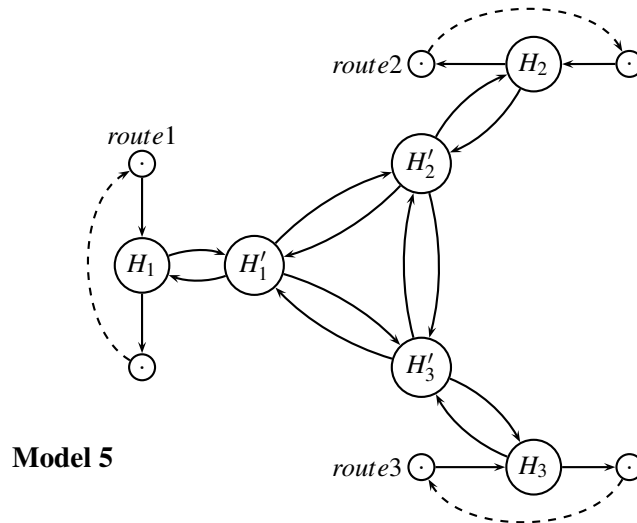
Note: The route net is the same as in Project Assignment 2019, except that Route 1 is changed so it arrives on a Wednesday instead of a Saturday. Since both route 1 and 2 arrive Wednesday, there is not sufficient time to transship containers from one vessel to the other, hence in this case transshipment takes 7 days.

Crane capacity: The crane capacity u for loading/unloading is 10 containers per day for routes 1 and 2, while the capacity u for loading/unloading is 7 containers per day for route 3. There are separate cranes for loading and unloading both having the stated capacity. Each vessel has its own two cranes, meaning that route 1 and route 2 do not share the capacity u . **Transshipment cost:** The cost of loading/unloading is 5\$ for route 1 and 2, while it is 3\$ for route 3. **Transshipment time:** The time for transferring a container from one vessel to another can be calculated from the time table. If two vessels arrive the same day, transshipment takes 7 days. **Storage capacity:** The port has a limited storage capacity of $\bar{u} = 15$ containers on shore. **Storage cost:** The shipping company must pay $\bar{c} = 2\$$ for each container each night it is stored in the port.

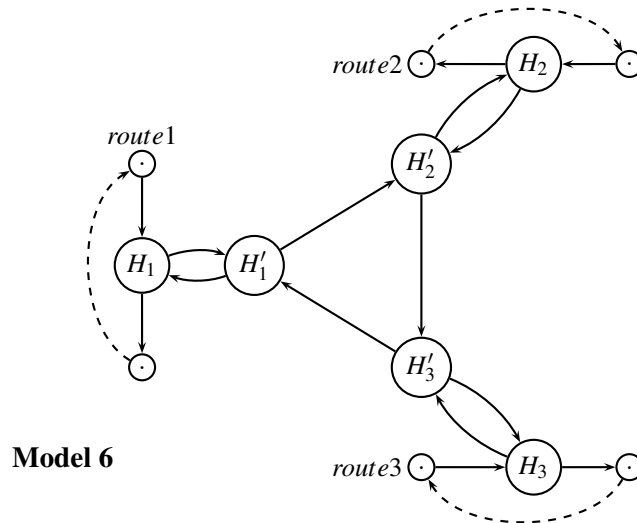
We assume that each vessel is only one day in the port, and it is assumed that all loading and unloading is done during this day. We now consider three different models for the hub port H as known from the project assignment 2019.



(Model 3) A possible way of modeling the hub port H is to use an extra node H_1 , H_2 and H_3 for each route. Edges are added between all pairs of nodes H_1 , H_2 , and H_3 .



(Model 5) Next, we may model the hub port H by using an extra node H_1 , H_2 and H_3 for each route, and additional nodes H'_1 , H'_2 and H'_3 . Edges are added between all pairs of nodes H'_1 , H'_2 , and H'_3 .



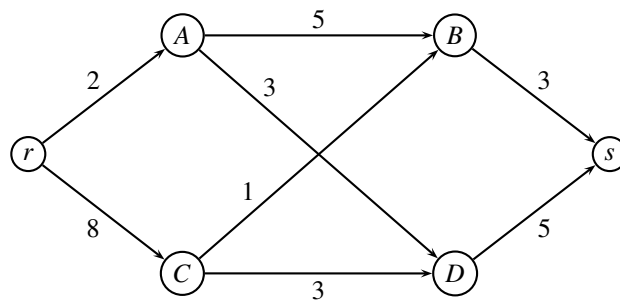
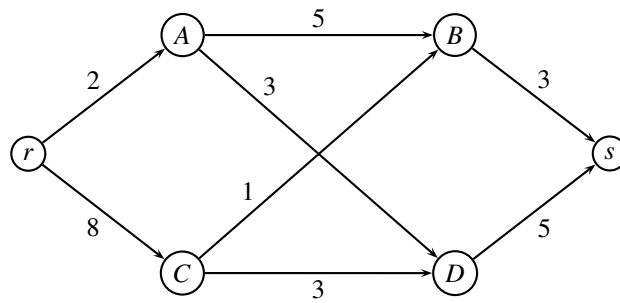
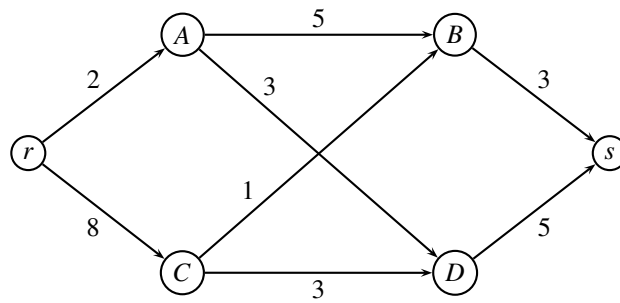
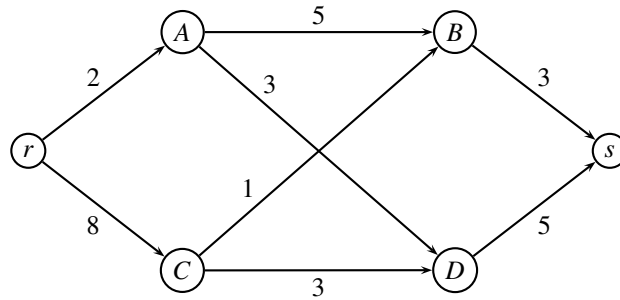
(Model 6) Finally, we may model the hub port H by using an extra node H_1 , H_2 and H_3 for each route, and additional nodes H'_1 , H'_2 and H'_3 . The latter nodes are connected in a cycle ordered according to the time of arrival.

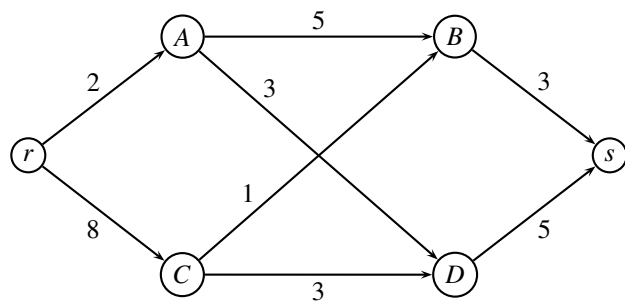
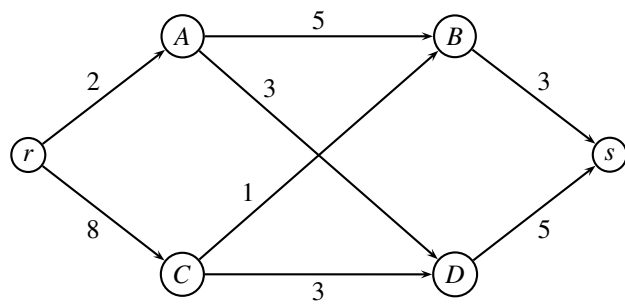
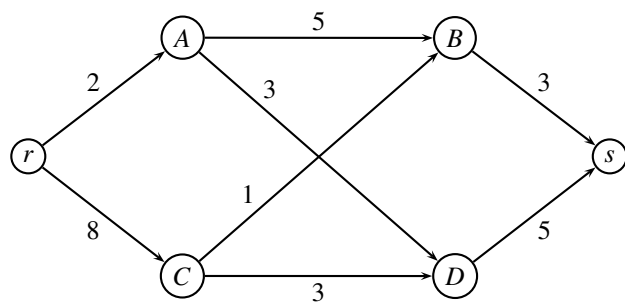
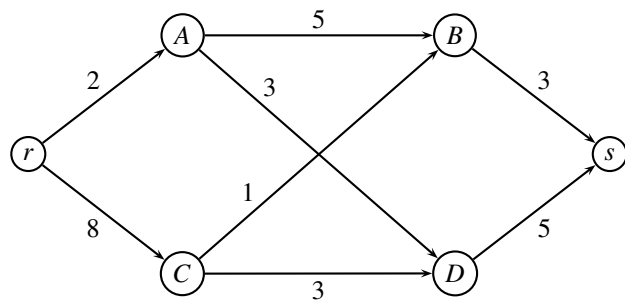
Problem 13: (5 point) Indicate on the models whether transshipment costs, transshipment time, crane capacity, storage capacity, and storage cost can be modeled as a multi-commodity flow problem using the three proposed models (Model 3, 5, 6). You can use the figures in the appendix. If the answer is positive, state the cost/capacity/time on the edges. If the answer is negative, argue why. ■

_____ end of assignment _____

Problem 6,7

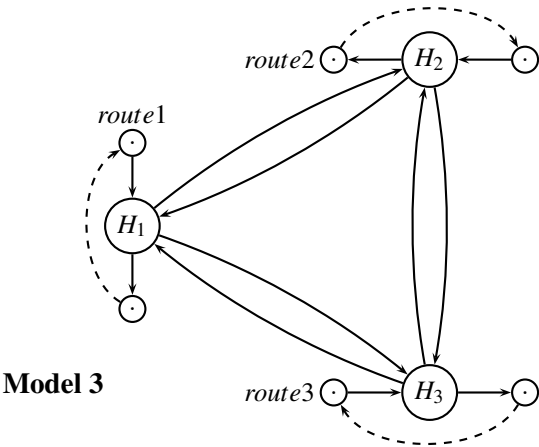
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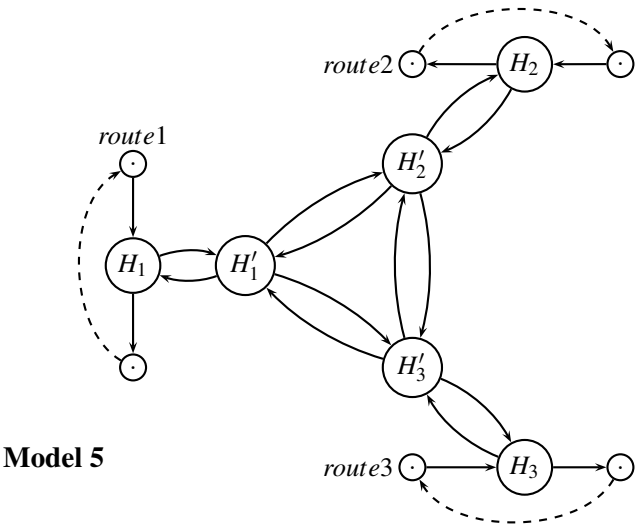


Problem 13

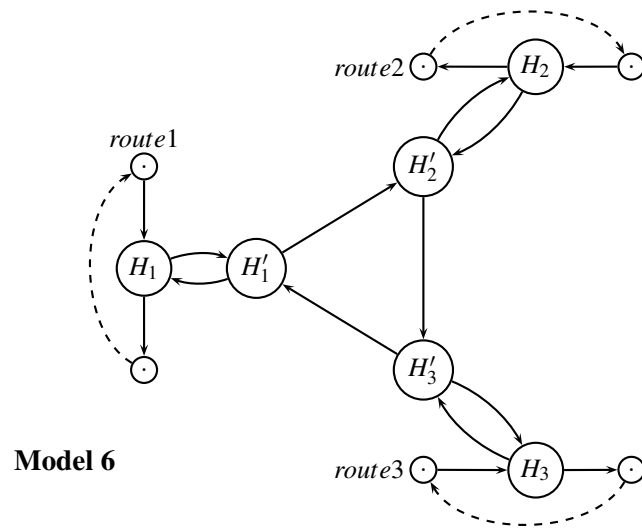
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Model	transshipment cost	transshipment time	crane capacity	storage capacity	storage cost
3					



Model	transshipment cost	transshipment time	crane capacity	storage capacity	storage cost
5					



Model	transshipment cost	transshipment time	crane capacity	storage capacity	storage cost
6					